H1 remediëring toets

zaterdag 4 oktober 2025 0:18

1) $\frac{144}{216} = \frac{2}{3}$

2) $\frac{46}{57} \cdot \frac{38}{69} = \frac{(460 \cdot 3)+(46 \cdot 8)}{(57 \cdot 60)+(57 \cdot 9)} = \frac{(1200+180)+(320+48)}{(3000+420)+(513)}$

$= \frac{(1380)+(368)}{(3420)+(513)} = \frac{1748}{3933} = \frac{92}{207} = \frac{4}{9}$

kgd = 19 :23

of

$\frac{46}{57} \cdot \frac{38}{69} = \frac{2 \cdot \cancel{23}}{3 \cdot \cancel{19}} \cdot \frac{2 \cdot \cancel{19}}{3 \cdot \cancel{23}} = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$

3) $\frac{\frac{2}{7}+\frac{5}{6}}{\frac{1}{5}+\frac{3}{4}} = \frac{\frac{12}{42}+\frac{35}{42}}{\frac{4}{20}+\frac{15}{20}} = \frac{\frac{47}{42}}{\frac{19}{20}} = \frac{47}{42} \cdot \frac{20}{19} = \frac{940}{798} = \frac{470}{399}$

4) $\frac{5}{8a} + \frac{1}{6b} + \frac{5}{12a^2b} = \frac{5 \cdot 3ab}{24a^2b} + \frac{1 \cdot 4a^2}{24a^2b} + \frac{5 \cdot 2}{24a^2b} = \frac{15ab + 4a^2 + 10}{24a^2b}$

5) a) $\left(\frac{-3}{5}\right)^{-2} = \frac{1}{\left(\frac{-3}{5}\right)^2} = \frac{1}{\frac{(-3)^2}{5^2}} = \frac{1}{\frac{9}{25}} = \frac{25}{9}$

b) $\left(-\frac{2}{3}\right)^{-3} = \frac{1}{\frac{(-2)^3}{3^3}} = \frac{1}{\frac{-8}{27}} = -\frac{27}{8}$

6) $\frac{y^{m-2}}{y^{m+1}} = \frac{1}{y^{m+1-(m-2)}} = \frac{1}{y^{m+1-m+2}} = \frac{1}{y^{1+2}} = \frac{1}{y^3}$

7)

a) $(y^{3+m})^2 = y^{6+m^2}$

b) $\left(\frac{-b^5}{b^4}\right)^3 = -\frac{b^{15}}{b^{12}} = -b^3$

b) $\left(\frac{-b^3}{b^4}\right) = -\frac{b^{15}}{b^{-12}} = -b^3$

8) $\frac{(a^2 b^{-7} c^{10})^3}{(a^3 b^4 c)^2 (a^{-2} b^{-3} c^3)^2} = \frac{\cancel{a^6} b^{-21} \cancel{c^{30}}}{(\cancel{a^6} b^8 \cancel{c^2})(a^{-4} b^{6} \cancel{c^6})} = \frac{a^4 \quad c^{22}}{b^{23}}$

9) $\frac{3 \cdot 10^{-8} \cdot 4 \cdot (10^{-2})^{\frac{1}{2}}}{6 \cdot (10^{-4} \cdot 10^{2})^{-3}} = \frac{3 \cdot 10^{-8} \cdot 4 \cdot 10^{-2 \cdot \frac{1}{2}}}{6 \cdot (10^{-2})^{-3}} = \frac{12 \cdot 10^{-9}}{6 \cdot 10^{6}} = 2 \cdot 10^{-15} = \frac{2}{10^{15}}$

10)

a) $\sqrt{242} = \sqrt{121 \times 2} = \sqrt{11^2 \times 2} = 11\sqrt{2}$

b) $\sqrt{441} = 21^2$

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a) $2\sqrt{14} \cdot (-3\sqrt{21}) = (2 \cdot -3)\sqrt{21 \cdot 14} = -6\sqrt{294} = -6\sqrt{49 \cdot 6} = -6\sqrt{7^2 \cdot 6}$

$= -6 \cdot 7\sqrt{6} = -42\sqrt{6}$

b) $-4\sqrt{22} \cdot 5\sqrt{33} = (-4 \cdot 5)\sqrt{22 \cdot 33} = -20\sqrt{726} = -20\sqrt{121 \times 6} = -20 \cdot 11\sqrt{6}$

$= -220\sqrt{6}$

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$-3\sqrt{30} \cdot 12\sqrt{14} \cdot (-\sqrt{21}) = (-3 \cdot 12 \cdot -1)\sqrt{30 \cdot 14 \cdot 21}$

$= 36\sqrt{8820} = (36 \cdot 2 \cdot 21)\sqrt{5} = 1512\sqrt{5}$

13

$-5\frac{\sqrt{3}}{\sqrt{15}} = -5 \cdot \frac{\sqrt{3}}{\sqrt{3}\sqrt{5}} = -5 \cdot \frac{1}{\sqrt{5}} = -\frac{5}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = -\frac{\cancel{5}\sqrt{5}}{\cancel{5}} = -\sqrt{5}$

14 $\sqrt[3]{200} \cdot \sqrt[3]{35} = \sqrt[3]{200 \cdot 35} = \sqrt[3]{7000} = \sqrt[3]{7 \cdot 1000} = \sqrt[3]{7 \cdot 10^3}$

$= 10 \sqrt[3]{7}$

15 $\dfrac{\sqrt[3]{-3}}{\sqrt[3]{2}} = -\dfrac{\sqrt[3]{3}}{\sqrt[3]{2}} \cdot \dfrac{\sqrt[3]{2^2}}{\sqrt[3]{2^2}} = -\dfrac{\sqrt[3]{3} \cdot \sqrt[3]{2^2}}{2} = -\dfrac{\sqrt[3]{3 \cdot 4}}{2} = -\dfrac{\sqrt[3]{12}}{2}$